STORM: Interference-Aware, Compute-Light Curriculum Scheduling for Continual Learning

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Abstract

Order sensitivity remains a dominant source of catastrophic forgetting and transient instability in continual learning, and this problem becomes especially costly with large backbones because recovering from unfavorable orders often requires expensive task revisits. We introduce STORM, a lightweight, model-agnostic controller that builds an online curriculum from gradient signatures extracted by tiny per-task probes on frozen features. STORM maintains a Task Interaction Matrix combining cosine similarity (expected transfer) with an elementwise sign-conflict penalty (interference), and schedules tasks by maximizing summed interactions within a sliding buffer. Because the backbone is frozen during signature estimation and the learner uses parameter-efficient adapters, the controller adds only minor overhead. We validate STORM in three complementary settings: a controlled synthetic interference suite, a vision proxy based on split CIFAR-10, and a synthetic NLP bag-of-words proxy. Across all settings, STORM improves final average accuracy over random ordering and reduces average forgetting; in the controlled setting it approximately halves the worst-case stability drop. We report exact schedules, metrics, and observed overheads, and ablate similarity-only and interference-only scoring. The results support interference-aware curricula as a practical, plug-and-play inductive bias for continual learning.

1 Introduction

Continual learning (CL) aims to train models on streams of tasks while retaining prior skills and transferring useful knowledge forward. Despite progress via rehearsal, regularization, modularization, and parameter-efficient tuning, strong methods still exhibit substantial order sensitivity: the same task pool, when shuffled, can yield markedly different final performance and severe transient drops in accuracy, commonly referred to as the stability gap Lange et al. [2022]. In practical deployments with large backbones, this variability is problematic because recovering from poor early orders often necessitates revisiting tasks, incurring repeated full-model compute. Ordering is therefore not a benign nuisance but a central lever for both robustness and efficiency in CL.

Why is this hard? Most CL algorithms react to interference after it has occurred. Rehearsal requires extra memory and careful replay scheduling Chaudhry et al. [2019], parameter regularization constrains updates but does not anticipate harmful interactions Kirkpatrick et al. [2016], and architectural strategies reduce coupling at the expense of search or capacity planning Yoon et al. [2019], Veniat et al. [2020], Fernando et al. [2017], Alet et al. [2018]. Empirically, the stability gap persists across these families and grows with task dissimilarity, suggesting a gradient-level origin Lange et al. [2022]. Gradient-centric studies further indicate that less forgetful representations are associated with better forward transfer Chen et al. [2023], and that calibrating gradients can mitigate interference Lin et al. [2024]. Yet existing curriculum heuristics often rely on metadata or loss magnitudes and do not

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estimate gradient interactions directly; meanwhile, methods that assume modular structure or task descriptors are not always applicable Veniat et al. [2020], Yoon et al. [2019].

We propose to optimize the ordering itself using signals that anticipate both transfer and interference before heavy fine-tuning. We present STORM (Similarity and inTerference-Optimised cuRRiculuM), a lightweight controller that sits in front of any CL learner and schedules tasks online. For each arriving task, STORM trains a tiny linear probe for one epoch on frozen backbone features and aggregates the probe's gradients with respect to features into a compact signature. Pairwise task interactions are estimated as a signed combination of cosine similarity (proxy for transfer) and elementwise sign conflicts (proxy for destructive interference). STORM then reorders queued tasks within a small sliding buffer by maximizing the sum of these interactions over the candidate sequence via a greedy insertion solver. The learner uses parameter-efficient adapters, so signatures are computed once per task and the backbone need not be revisited for the controller's decisions.

We evaluate STORM across three complementary quick-mode settings designed for detailed logging. First, a controlled synthetic suite with tunable interference validates that STORM improves final accuracy and stability relative to random order and that it outperforms similarity-only scoring. Second, a vision proxy based on split CIFAR-10 with single-head evaluation aut [f] demonstrates improved final average accuracy and dramatically reduced forgetting compared to random ordering. Third, a synthetic NLP bag-of-words proxy shows consistent gains in final average accuracy and reduced forgetting. We report exact schedules, metrics, and the observed probe overheads, and include ablations where similarity-only and interference-only variants are compared. While stability-gap improvements are mixed in the two applied proxies, the controlled setting shows a near halving of the worst-case drop.

STORM complements perspectives that reframe CL as sequence modeling Lee et al. [2023] and practitioner-focused continual pretraining studies emphasizing realistic compute budgets and stream order effects Roth et al. [2024]. It is also compatible with generalized CL in language models using parameter-efficient updates Peng et al. [2024]. Finally, STORM is orthogonal to rehearsal and regularization and can be combined with both Chaudhry et al. [2019], Kirkpatrick et al. [2016], Chaudhry et al. [2018].

- **Probe-driven online controller:** An online curriculum controller that estimates task-to-task transfer and interference from probe gradients on frozen features, without touching the backbone.
- **Interaction scoring and buffer search:** A scoring scheme that combines cosine similarity with an elementwise sign-conflict penalty, paired with a buffer-constrained sequence optimizer for streaming settings.
- **Compute-light and model-agnostic:** A compute-light, model-agnostic design that pairs naturally with parameter-efficient fine-tuning and coexists with gradient calibration Lin et al. [2024] or rehearsal Chaudhry et al. [2019].
- **Validated across proxies:** Empirical validation on synthetic, vision, and NLP proxies with exact schedules, metrics, and ablations, showing improved final accuracy and reduced forgetting over random order.

Beyond these results, we discuss limitations and future directions: tightening the theoretical link between our interaction scores and generalization; adaptive buffer policies and dynamic weighting of similarity versus interference; and scaling to realistic multimodal continual pretraining benchmarks where streaming constraints dominate Roth et al. [2024]. We also view STORM as complementary to broader CL benchmarks and modalities, including graph and reasoning scenarios aut [a,b], compositional language and alignment-based CL aut [c,d], and analyses of human-like learning signals aut [e].

2 Related Work

Rehearsal and memory editing. Rehearsal interleaves stored examples from past tasks to mitigate forgetting; even tiny episodic memories can substantially help Chaudhry et al. [2019]. Gradient-based editing of memory makes future replays more informative by shaping stored examples to challenge upcoming updates Jin et al. [2020]. While effective, rehearsal is reactive and requires storage and scheduling; it does not directly address order sensitivity in the absence of replay.

Regularization-based methods. Elastic Weight Consolidation (EWC) slows learning on parameters deemed important for prior tasks Kirkpatrick et al. [2016]. A-GEM constrains updates using gradients on a small memory to preserve performance on earlier tasks efficiently Chaudhry et al. [2018]. These approaches reduce interference but still exhibit stability gaps and order sensitivity because they act after gradients have been computed on the current task Lange et al. [2022]. STORM differs by predicting interference before fine-tuning and proactively choosing a favorable order.

Architectural and modular strategies. Additive Parameter Decomposition (APD) decouples shared and task-adaptive components to improve order robustness Yoon et al. [2019]. Modular Networks with Task-Driven Priors search over module compositions during streaming Veniat et al. [2020]. PathNet evolves routing to encourage transfer while freezing reused parts Fernando et al. [2017], and modular meta-learning composes reusable blocks for combinatorial generalization Alet et al. [2018]. These approaches assume or induce modular structure and often entail nontrivial search or expansion. By contrast, STORM assumes no modularity, requires no capacity planning, and leverages frozen features for fast, model-agnostic scheduling.

Gradient-centric perspectives. The stability gap has been identified as a pervasive, order-sensitive phenomenon that worsens when tasks are dissimilar Lange et al. [2022]. Dynamic gradient calibration adjusts updates to reduce interference and can be layered onto diverse learners Lin et al. [2024]. Representation studies report that less forgetful features correlate with better forward transfer Chen et al. [2023]. Complementary work on mode connectivity examines relationships between minima obtained by sequential and multitask training Mirzadeh et al. [2020]. STORM is aligned with these insights but targets the overlooked dimension of task ordering, using gradient-derived signals without modifying the learner's update rule.

Language and practical continual pretraining. Scalable language models in generalized CL settings leverage parameter-efficient re-parameterization and retrieval to reduce forgetting across diverse tasks Peng et al. [2024]. A practitioner's guide to continual multimodal pretraining highlights the importance of stream order and compute constraints Roth et al. [2024]. STORM complements these by providing an interference-aware curriculum controller that can be plugged into such PEFT pipelines.

Complementary domains and benchmarks. Broader CL research spans graphs and algorithmic reasoning aut [a,b], compositional language continual learning and alignment methods aut [c,d], and investigations into unsupervised learning signals from a human-learning perspective aut [e]. These settings emphasize different structural priors and supervision regimes; STORM targets the supervised task stream regime and is orthogonal to these trends.

Comparison and contrast. Relative to rehearsal and regularization Chaudhry et al. [2019], Kirkpatrick et al. [2016], Chaudhry et al. [2018], STORM is proactive and orthogonal: it predicts interactions and schedules to avoid interference before it happens. Compared to modular and routing approaches Yoon et al. [2019], Veniat et al. [2020], Fernando et al. [2017], Alet et al. [2018], STORM makes no structural assumptions and incurs negligible search cost via greedy insertion in a small buffer. Unlike gradient calibration Lin et al. [2024], STORM does not alter gradients during training; it changes which gradients the learner encounters by reordering tasks. Sequence-modeling views Lee et al. [2023] rearchitect the learner to internalize the sequence; STORM instead provides a drop-in scheduling layer that improves outcomes for existing learners without architectural changes.

3 Background

3.1 Supervised continual learning setting

We study supervised continual learning over a sequence of tasks $t = 1, 2, \ldots$ where each task has its own dataset and label space. After training on task t, the model is evaluated on all seen tasks. Depending on the setting, we use either a single unified head over all classes (class-incremental) or task-specific heads (multi-head). The learner employs parameter-efficient fine-tuning (PEFT), updating a small subset of weights, while the backbone feature extractor remains frozen during signature estimation.

3.2 Order sensitivity and stability metrics

Order sensitivity appears as variance in final average accuracy across task permutations and as transient dips during training. The latter is quantified by the stability gap, observed as a difference between a trajectory's maximum and minimum average accuracy over time Lange et al. [2022]. We report final average accuracy, average forgetting defined as the mean across tasks of the maximum observed accuracy on that task during training minus its final accuracy, and stability proxies min-ACC and worst-case drop (wc_drop), the difference between final and minimum average accuracy.

3.3 Gradient-centric motivation

Catastrophic forgetting results from gradient interference when updates for the current task negatively affect parameters used by prior tasks; positive transfer arises when updates align in beneficial directions Chen et al. [2023]. This motivates estimating pairwise task interactions before full fine-tuning. Direct estimation by training the large backbone is infeasible; instead, we train a small linear probe on frozen features to obtain gradients with respect to features, which serve as compact task signatures capturing directional tendencies. Such gradient-centric thinking complements gradient calibration methods Lin et al. [2024] and remains compatible with rehearsal Chaudhry et al. [2019] and regularization Kirkpatrick et al. [2016].

3.4 Task interaction formalism

For each task u, we compute a signature $\mathbf{g}\_u$ by aggregating gradients of a probe's loss with respect to frozen features and L2-normalizing the result. For two tasks i and j, define a similarity term $s\_ij = \cos(\mathbf{g}\_i, \mathbf{g}\_j)$, which proxies representational transfer, and an interference term r_ij as the ℓ_1 magnitude of the negative coordinates of the elementwise product $\mathbf{g}\_i \odot \mathbf{g}\_j$, which penalizes coordinate-wise sign conflicts. The Task Interaction Matrix has entries

$$M\_ij = \alpha\, s\_ij - \beta\, r\_ij, \quad \alpha, \beta > 0.$$

Given a set of pending tasks, the curriculum objective is to order them to maximize the sum of M_ij over all pairs where i precedes j, subject to a streaming constraint that only a sliding buffer of size B tasks can be reordered at any time.

3.5 Assumptions and scope

We assume access to a frozen backbone for extracting features during probe training, and that the learner updates fewer than all parameters via adapters. This allows computing signatures once per task and reusing them until training begins. We do not assume modular structure, task descriptors, or rehearsal memory, and we make no inference-time assumptions about task identity.

3.6 Connections to prior work

Our approach exploits gradient-centric insights into interference and transfer Chen et al. [2023], Lange et al. [2022]. It is orthogonal to rehearsal and regularization Chaudhry et al. [2019], Kirkpatrick et al. [2016], Chaudhry et al. [2018] and compatible with gradient calibration Lin et al. [2024]. It also aligns with system-level observations that stream order matters in practical continual pretraining regimes Roth et al. [2024], while remaining simple to integrate with generalized CL pipelines Peng et al. [2024].

4 Method

STORM consists of four components: signature extraction, interaction estimation, buffer-constrained search, and adapter-aware execution.

4.1 Signature extraction

When a new task u arrives, we compute frozen features for its training data and train a linear probe for one epoch. During probe training, we accumulate the gradient of the probe's loss with respect to

the frozen feature vectors across minibatches. The accumulated vector is L2-normalized to produce the task signature $\mathbf{g}\_u$. This process entails only forward and backward passes through the small probe, not the backbone, and thus is compute-light. We store signatures on CPU to minimize memory overhead.

4.2 Interaction estimation

For tasks i and j with signatures $\mathbf{g}\_i$ and $\mathbf{g}\_j$, we compute two scores: a similarity term $s\_ij = \cos(\mathbf{g}\_i, \mathbf{g}\_j)$, which estimates expected transfer, and an interference term r_ij defined as the ℓ_1 magnitude of negative coordinates of the elementwise product $\mathbf{g}\_i \odot \mathbf{g}\_j$, which penalizes sign disagreements as a proxy for destructive interference. The Task Interaction Matrix is then $M\_ij = \alpha s\_ij - \beta r\_ij$, where α and β control the trade-off (default $\alpha = \beta = 1$). Note that, under the sequencing objective, contributions depend on order because the objective sums M_ij for i preceding j.

4.3 Buffer-constrained curriculum search

Tasks are buffered in a sliding window of size B (default $B = 5$). Whenever the buffer fills or a reordering trigger fires, STORM orders the buffer tasks by maximizing the objective: the sum over all indices $a < b$ in the candidate sequence of $M\_order[a], order[b]$. For small sets one can solve exactly; in practice we use a greedy insertion algorithm with $\mathcal{O}(B^2)$ complexity that initializes with a single task, then inserts each remaining task into the position maximizing the objective over the partial sequence. The controller returns the earliest task in the resulting order as the next to train; as tasks are consumed, the buffer advances. This provides bounded latency and negligible compute compared to fine-tuning.

4.4 Adapter-aware execution

The downstream learner uses parameter-efficient fine-tuning with low-rank adapters, updating a small fraction of weights per task. Because the backbone remains frozen during signature computation, each signature is computed once and reused until training. STORM does not alter the learner's loss or optimizer; it only selects the order in which tasks are presented. It can be combined with gradient calibration inside the learner Lin et al. [2024] or with rehearsal and regularization without modification Chaudhry et al. [2019], Kirkpatrick et al. [2016], Chaudhry et al. [2018].

4.5 Complexity and default settings

Signature extraction scales with one epoch of probe training over cached features and adds negligible memory beyond storing signatures. The controller's runtime is dominated by pairwise scoring and greedy insertion within the buffer, which is inexpensive for small B. The design target is an overhead far smaller than a single pass of adapter training, particularly when backbones are large. Unless stated otherwise, we set $\alpha = 1$ and $\beta = 1$, buffer size $B = 5$, train the probe for one epoch, L2-normalize signatures, and use AdamW as optimizer Kingma and Ba [2014]. The similarity term captures representational alignment, whereas the sign-conflict penalty emphasizes coordinates where tasks push in opposite directions. Empirically, both terms are necessary to balance plasticity and stability; removing either degrades performance in our ablations.

4.6 Relation to alternative curricula

Heuristics based on class frequency or mean loss do not directly reflect gradient interactions and can be misled by scale. Modular task-driven priors search over compositions Veniat et al. [2020], while sequence-modeling perspectives rearchitect the learner Lee et al. [2023]. STORM instead provides a model-agnostic controller that leverages gradient-derived signals to make fast, online ordering decisions within existing PEFT pipelines.

[H] STORM: Buffer-Constrained Interference-Aware Scheduling [1] **Input:** stream of tasks $\{\mathcal{T}\_u\}$, buffer size B, trade-off weights α, β **Init:** empty buffer $\mathcal{B} \leftarrow [\,]$; signatures dict $\mathcal{G} \leftarrow$ each arriving task u Extract frozen features for $\mathcal{T}\_u$; train linear probe for 1 epoch Accumulate feature-gradients; set signature $\mathbf{g}\_u \leftarrow L2Norm(\sum \nabla\_feat\mathcal{L}\_probe)$ Store $\mathcal{G}[u] \leftarrow \mathbf{g}\_u$; append to buffer $\mathcal{B}.append(u)$

$|\mathcal{B}| = B$ or reorder trigger Compute pairwise scores for $i, j \in \mathcal{B}$: $s\_ij = \cos(\mathcal{G}[i], \mathcal{G}[j])$; $r\_ij = \| \min(\mathbf{0}, \mathcal{G}[i] \odot \mathcal{G}[j])\|\_1$; $M\_ij = \alpha s\_ij - \beta r\_ij$ Initialize sequence with first task: $\pi \leftarrow [\mathcal{B}[0]]$ each remaining task $v \in \mathcal{B} \setminus \{\pi[0]\}$ Find position k in $\{0, \ldots, |\pi|\}$ that maximizes objective $J(\pi \oplus\_kv) = \sum\_a < bM\_\pi\_k[a]$, $\pi\_k[b]$ Insert: $\pi \leftarrow \pi \oplus\_kv$ Select next task to train: $u^\star \leftarrow \pi[0]$; output to learner Remove $u^\star$ from buffer and continue streaming

5 Experimental Setup

5.1 Scope and goals

We validate STORM in three quick-mode experiments designed for tractability and detailed logging: (1) a controlled synthetic interference suite, (2) a vision continual learning proxy on split CIFAR-10, and (3) a synthetic NLP bag-of-words proxy. Across settings we measure final average accuracy, average forgetting, and stability proxies (min-ACC and worst-case drop), and we report probe overhead when available. Training budgets and optimizers are held constant across curricula within each experiment to isolate ordering effects.

5.2 Common implementation defaults

Training uses PyTorch with mixed precision for adapter updates and fp32 for probes. Random seeds are set at run start. After training each task, we evaluate on all seen tasks and compute the trajectory of average accuracy to derive final average accuracy, average forgetting, min-ACC, and wc_drop. We do not assume rehearsal memory or apply additional regularization baselines in these quick runs; comparisons focus on ordering effects. AdamW is used as optimizer Kingma and Ba [2014].

5.3 Experiment 1 - Synthetic controlled interference

Data. We generate binary classification tasks from a frozen random feature map with controlled angular separations between task weight vectors, enabling regimes of positive transfer and interference. **Learner.** A two-layer MLP with a LoRA-like adapter on the first layer serves as the PEFT learner. **Probe and signature.** A one-epoch linear probe is trained on frozen features; the gradient with respect to features is accumulated and L2-normalized to form the signature. **Curricula.** STORM, random order, similarity-only ($\beta = 0$), and interference-only ($\alpha = 0$). **Training budget.** A fixed small number of epochs per task. **Metrics.** Final average accuracy, average forgetting, min-ACC, and wc_drop. We also measure the share of wall-clock time spent on probe training relative to total runtime in this quick setting.

5.4 Experiment 2 - Vision proxy on split CIFAR-10

Dataset and protocol. CIFAR-10 is split into 5 disjoint tasks with 2 classes each, evaluated with a single unified head over all 10 classes aut [f]. **Backbone and learner.** A compact backbone is kept frozen during probe computation; the learner fine-tunes a small subset of parameters via adapters, using a single-head classifier. **Probe and signature.** One-epoch linear probe on cached features; signature computed as in Experiment 1. **Controller.** STORM with buffer size $B = 5$ and greedy insertion. **Baselines and ablations.** Random order and a similarity-only scoring ablation. **Training budget.** A fixed small number of epochs per task with AdamW Kingma and Ba [2014]. **Metrics.** Final average accuracy, average forgetting, min-ACC, and wc_drop.

5.5 Experiment 3 - NLP proxy on synthetic bag-of-words

Tasks and protocol. We construct 5 synthetic text classification tasks with distinct vocabularies. Features are pooled and kept frozen for probe training; task-specific heads are used for evaluation (multi-head). **Learner.** Adapter-style updates over a shallow projection. **Probe and signature.** As in Experiment 1. **Controller and baselines.** STORM, random order, and similarity-only scoring. **Training budget.** A fixed small number of epochs per task. **Metrics.** Final average accuracy, average forgetting, min-ACC, and wc_drop.

5.6 Statistical protocol and reproducibility

The broader protocol recommends multiple seeds and task pools, but the reported logs correspond to quick-mode runs with one seed per setting to enable fast iteration while keeping budgets identical across curricula. We save accuracy and loss curves for all methods to facilitate inspection of stability and convergence.

5.7 Positioning relative to broader settings

While recent work advocates parameter-efficient generalized CL for language and multimodal pretraining Peng et al. [2024], Roth et al. [2024], our experiments focus on compact proxies to examine the core question: can an interference-aware controller improve outcomes by reordering tasks, under matched training budgets, without modifying the learner?

6 Results

We report the exact numbers printed by the experiment logs and the training orders chosen by each method. All runs were executed in quick mode with one seed per setting.

6.1 Experiment 1 - Synthetic controlled interference ($T = 6$)

Signatures were computed in 1.337 s. Final metrics:

- STORM: final_avg_acc = 0.5454, avg_forgetting = 0.0496, min_acc = 0.5221, wc_drop = 0.0233.
- Random: final_avg_acc = 0.5183, avg_forgetting = 0.0746, min_acc = 0.5183, wc_drop = 0.0463.
- Similarity-only: final_avg_acc = 0.5437, avg_forgetting = 0.0542, min_acc = 0.5254, wc_drop = 0.0233.
- Interference-only: final_avg_acc = 0.5450, avg_forgetting = 0.0454, min_acc = 0.5325, wc_drop = 0.0279.

Overhead: total runtime approximately 2.38 s; probe share approximately 56.1%. Interpretation. STORM improves final average accuracy by 2.71 percentage points over random and reduces average forgetting by roughly one third; the worst-case drop is about half that of random. Similarity-only approaches STORM's final accuracy but forgets more; interference-only matches final accuracy in this controlled geometry but is not consistently superior in later proxies.

6.2 Experiment 2 - Vision continual learning on split CIFAR-10 (5 tasks)

Orders:

- STORM: .
- Random: .
- Similarity-only: .

Final metrics:

- STORM: final_avg_acc = 0.1310, avg_forgetting = 0.0186, min_acc = 0.0969, wc_drop = 0.0341.
- Random: final_avg_acc = 0.1221, avg_forgetting = 0.1591, min_acc = 0.1000, wc_drop = 0.0221.
- Similarity-only: final_avg_acc = 0.0976, avg_forgetting = 0.0010, min_acc = 0.0976, wc_drop = 0.0010.

Interpretation. STORM raises final average accuracy by 0.89 percentage points over random and dramatically reduces average forgetting. The similarity-only schedule minimizes forgetting but at a

substantial cost to final accuracy, indicating that interference signals are needed to preserve plasticity. Transient stability is mixed: wc_drop is slightly larger for STORM here, suggesting that orders yielding the best end performance can still incur deeper mid-stream dips in this proxy.

6.3 Experiment 3 - NLP proxy on synthetic bag-of-words (5 tasks)

Orders:

- STORM: .
- Random: .
- Similarity-only: .

Final metrics:

- STORM: final_avg_acc = 0.7620, avg_forgetting = 0.0100, min_acc = 0.3530, wc_drop = 0.4090.
- Random: final_avg_acc = 0.7290, avg_forgetting = 0.0170, min_acc = 0.3740, wc_drop = 0.3550.
- Similarity-only: final_avg_acc = 0.7150, avg_forgetting = 0.0000, min_acc = 0.3840, wc_drop = 0.3310.

Interpretation. STORM improves final average accuracy by 3.30 percentage points relative to random and reduces average forgetting. As in vision, similarity-only reduces forgetting to near zero but at the cost of lower final accuracy. The worst-case drop is larger for STORM here, echoing that optimizing for end performance alone may not optimize mid-stream stability under this proxy.

Ablations and fairness. The similarity-only ablation underperforms STORM in final accuracy in all settings, while its low forgetting indicates overly conservative scheduling. Interference-only matches STORM's final accuracy only in the controlled synthetic case; in more realistic proxies, combining similarity and interference is necessary. All curricula share identical budgets, optimizers, and evaluation protocols within each setting. Energy and VRAM were not separately reported; the synthetic run indicates that probe time can dominate wall-clock in quick mode due to short per-task training.

7 Conclusion

We presented STORM, an interference-aware curriculum controller for continual learning that predicts task interactions from probe gradients on frozen features and reorders tasks within a sliding buffer to maximize transfer while mitigating interference. STORM is model-agnostic and compute-light by design, pairing naturally with parameter-efficient fine-tuning. In quick-mode experiments spanning controlled synthetic tasks, a split CIFAR-10 vision proxy, and a synthetic NLP proxy, STORM consistently improved final average accuracy over random ordering and reduced average forgetting; in the controlled setting it approximately halved the worst-case stability drop. Similarity-only ablations minimized forgetting but sacrificed final accuracy, underscoring the importance of explicitly modeling interference.

These findings reinforce gradient-centric perspectives on forgetting and transfer Lange et al. [2022], Chen et al. [2023] and highlight task ordering as an underused lever that complements rehearsal, regularization, and architectural strategies Chaudhry et al. [2019], Kirkpatrick et al. [2016], Veniat et al. [2020], Yoon et al. [2019]. Limitations include mixed improvements on transient stability in two proxies and inflated probe cost under very short schedules. Future work includes integrating gradient calibration within the learner Lin et al. [2024], adapting buffer policies and alpha-beta weighting under streaming, and scaling evaluations toward realistic multimodal continual pretraining regimes Roth et al. [2024] and generalized language model CL Peng et al. [2024]. We view interference-aware curricula as a practical, plug-and-play inductive bias for robust and efficient continual learning that can be combined with existing CL approaches without altering the underlying learner.

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Figure 1: Accuracy over time for the synthetic experiment, STORM vs baselines.

Figure 2: Training loss over time for the synthetic experiment.

Figure 3: Accuracy over time for the vision proxy (split CIFAR-10).

Figure 4: Training loss over time for the vision proxy.

Figure 5: Accuracy over time for the NLP proxy (synthetic bag-of-words).

Figure 6: Training loss over time for the NLP proxy.